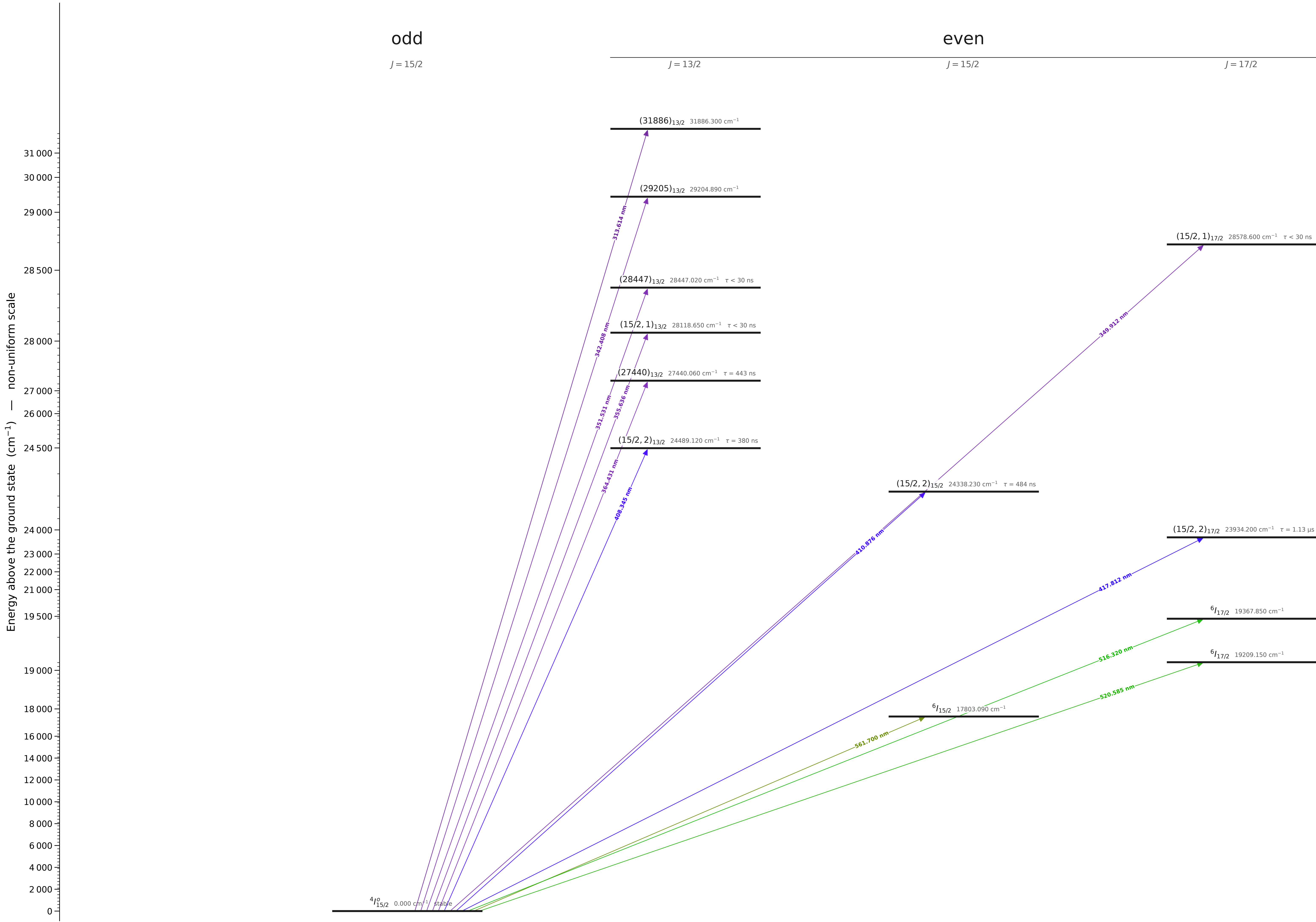


Einsteinium

Es (Es I). Classified transitions below 2 μm (levels and lines from the literature): 13 levels, 12 lines.

Es⁺ ionisation limit 51364.58 cm^{−1} = 6.36840 eV (194.687 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
⁴ f _{15/2} – (15/2, 1) _{17/2}	349.9122	349.8121	856.7649	–	–	5.305	–	first step of Mainz resonance ionization scheme No. 10
⁴ f _{15/2} – (28447) _{13/2}	351.5307	351.4302	852.8202	–	–	5.305	–	the einsteinium laser-spectroscopy line
⁴ f _{15/2} – (15/2, 1) _{13/2}	355.6359	355.5343	842.9759	–	–	5.305	–	first step of Mainz resonance ionization scheme No. 8
⁴ f _{15/2} – (27440) _{13/2}	364.4307	364.3269	822.6323	–	–	0.3593	–	Mainz resonance ionization scheme No. 7
⁴ f _{15/2} – (15/2, 2) _{13/2}	408.3446	408.2293	734.1653	–	–	0.4188	–	Mainz resonance ionization scheme No. 3
⁴ f _{15/2} – (15/2, 2) _{15/2}	410.8762	410.7603	729.6418	–	–	0.3288	–	Mainz resonance ionization scheme No. 2
⁴ f _{15/2} – (15/2, 2) _{17/2}	417.8122	417.6944	717.5293	–	–	0.1405	–	Mainz resonance ionization scheme No. 1

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁵³Es hyperfine structure (*I* = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)									
⁴ f _{15/2}	817.153(7)	-4316.25(8)	F=4: -25882.65	F=5: -21004.10	F=6: -15343.64	F=7: -9006.96	F=8: -2117.39	F=9: +5184.13	F=10: +12739.06	F=11: +20371.20		
⁶ f _{15/2}	2047.400	-2377.400	F=4: -61776.20	F=5: -51102.54	F=6: -38400.88	F=7: -23729.45	F=8: -7156.18	F=9: +11241.31	F=10: +31375.68	F=11: +53149.90		
⁶ i _{17/2}	2816.700	-6514.500	F=5: -95930.56	F=6: -77798.63	F=7: -56956.06	F=8: -33546.56	F=9: -7734.36	F=10: +20295.78	F=11: +50338.57	F=12: +82168.20		
⁶ f _{17/2}	-113.200	42.000	F=5: +3778.57	F=6: +3091.43	F=7: +2291.77	F=8: +1380.52	F=9: +358.75	F=10: -772.37	F=11: -2011.51	F=12: -3357.20		
(15/2, 2) _{17/2}	2284(15)	–	F=5: -75943.00	F=6: -62239.00	F=7: -46251.00	F=8: -27979.00	F=9: -7423.00	F=10: +15417.00	F=11: +40541.00	F=12: +67949.00		
(15/2, 2) _{15/2}	1532(15)	–	F=4: -45577.00	F=5: -37917.00	F=6: -28725.00	F=7: -18001.00	F=8: -5745.00	F=9: +8043.00	F=10: +23363.00	F=11: +40215.00		
(15/2, 2) _{13/2}	99(15)	–	F=3: -2598.75	F=4: -2202.75	F=5: -1707.75	F=6: -1113.75	F=7: -420.75	F=8: +371.25	F=9: +1262.25	F=10: +2252.25		
(15/2, 1) _{13/2}	1303.400	-866.400	F=3: -34547.48	F=4: -29181.55	F=5: -22509.83	F=6: -14556.14	F=7: -5349.03	F=8: +5078.17	F=9: +16687.38	F=10: +29435.75		
(28447) _{13/2}	2766(21)	-4425(530)	F=3: -74309.42	F=4: -62467.40	F=5: -47847.22	F=6: -30570.45	F=7: -10782.97	F=8: +11345.03	F=9: +35619.04	F=10: +61820.25		
(29205) _{13/2}	1911.500	-2287.400	F=3: -51056.64	F=4: -43008.46	F=5: -33042.50	F=6: -21221.59	F=7: -7621.15	F=8: +7670.85	F=9: +24553.86	F=10: +42914.78		

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: energy (from the cited literature; NIST lists no excited levels) and measured lifetime (where one exists).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁴ f _{15/2} – (28447) _{13/2}	351.531	254-253	-5020(54)
⁴ f _{15/2} – (28447) _{13/2}	351.531	255-253	-11046(86)

Shift = ν(first isotope) – ν(second isotope).